

AWI-ESM3 ↔ PISM moving-cavity coupling

the investigation report

Single living document. Superseded claims are struck through, not deleted.

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Abstract

Mission. A working end-to-end AWI-ESM3 (FESOM2–CORE3-cavity + OIFS-48r1 + LPJ-GUESS + WAM) ↔ Antarctic-PISM cycle: awiesm3 year → PISM decade → awiesm3 year on a changed mesh, with all three coastline-following members (FESOM cavity, OIFS land–sea mask, WAM sea points) regenerated per leg.

Where we are. Seven real bugs and one architectural redesign (the coupled slab) later, the cycle is proven end-to-end with real cavities — awiesm3 year → PISM decade → awiesm3 year on the changed mesh, plus one further clean cycle. The 10-year production attempt then ran eleven more increments *with the cavities accidentally amputated by an overreaching mesh-hygiene fix* (discovered at wrap-up, reverted; those legs are tainted). The campaign continues, and the diagnosis has turned to physics: the Ross-front blowup is increment magnitude, not coastline fragility, and the increments are large because the PI-control cavity melt runs $\sim 7\times$ observations. That melt is traced not to the melt scheme or the ocean temperature but to cavity currents $\sim 6\times$ too fast; the plan is a separate, tunable ice-base drag coefficient (§1.11, §1.12).

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1 The campaign, in the order it happened

Each subsection carries the date it was worked. The bug numbering (I–V) is the order of *discovery* and is kept as the canonical naming.

1.1 The restart seam and the cold-start control (2026-07-08–13)

The original FESOM cavity-margin blowup: a remapped warm restart mixes an evolved, balanced dynamic state (98% of nodes, bit-exact) with patched values at changed nodes; the *seam* between them is the instability. The decisive control:

Run	Δt	Initial state	Outcome
full-mesh control	1200 s	native spun-up restart	stable, worst CFLz 2.61
PISM cavity, cold	1200 s	cold start	stable, CFLz 3.44
PISM cavity, warm	1200 s	remapped restart	dies day 4.0 (CFLz 9.2)
PISM cavity, warm	60 s	remapped restart	dies day 4.4 (η_n , 0 CFLz)

The ocean lives happily in PISM’s cavity; the production answer became: run chunk-1 cold-started on the PISM cavity (artifacts pooled), and let `remap_restart` carry only genuine per-leg increments. The first real increment test crashed on 17 fully-opened columns (ice removed entirely, 16 levels at once) out of 1399 — but its numbers are **TAINTED** (measured under Bugs I+II) and the finding that matters survived into the fixes below: absorbing 99% of a geometry change is not a passing grade when the remaining 1% is fatal.

1.2 Bug I — PISM was fed surface water as ice-base forcing (2026-07-14)

`coupling_ocean2pism` built PISM’s `-ocean` th forcing as a mean over FESOM level *indices* 1–20 = **0–410 m including the summer surface** (the correct `-sellevel,150/600` sat in dead code). PISM received ice-base water up to +6.5°C even from a healthy ocean, melted at ~30 m/yr, and shed **24% of its floating area per decade** — every mesh-change leg then had to absorb a catastrophic geometry increment. FESOM’s own cavity melt in the same runs was sane (2.2 m/yr mean, 0.72 median).

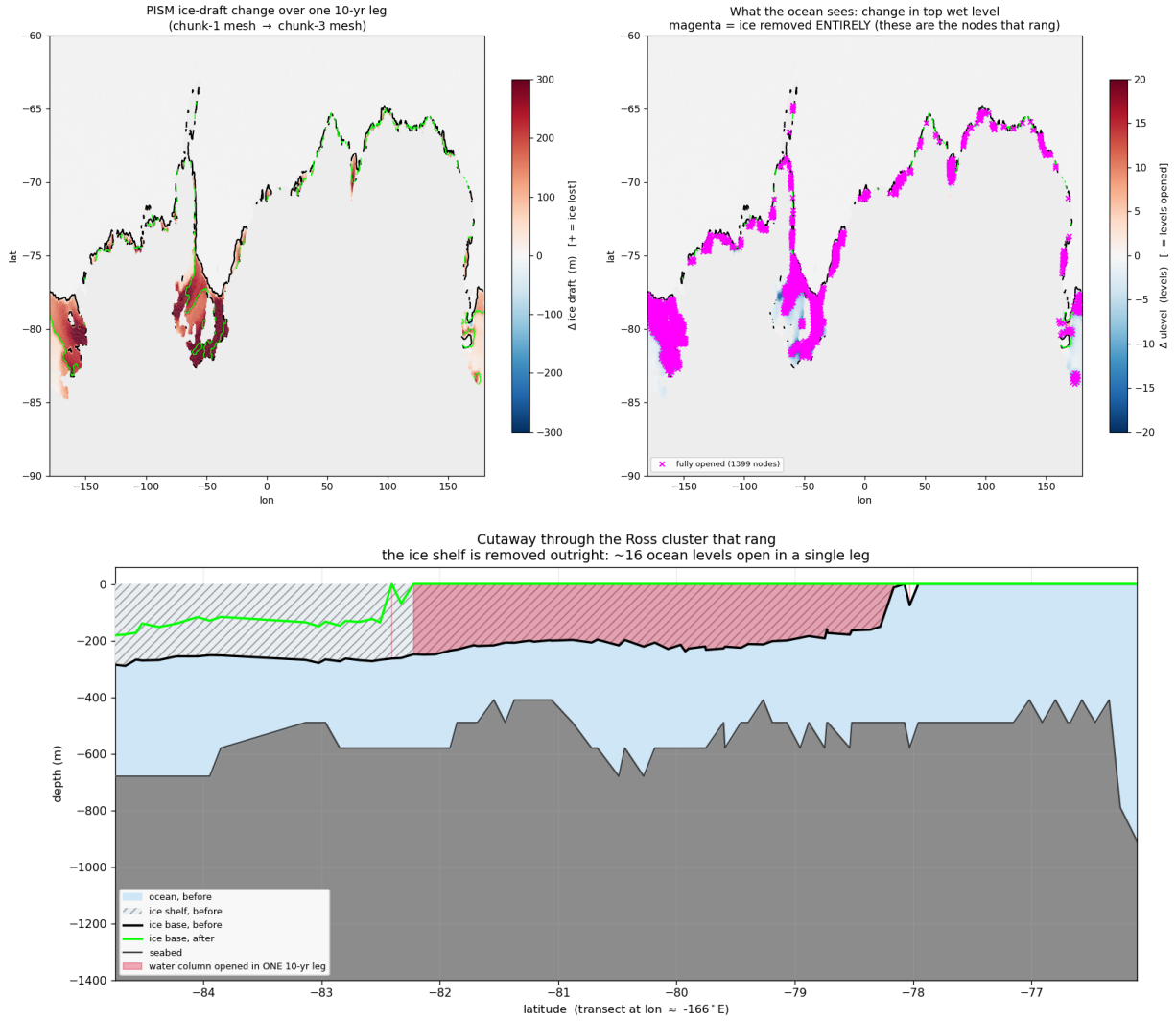


Figure 1: What Bug I did to PISM in one 10-year leg under surface-water forcing: 1399 nodes lose their ice entirely (magenta ring around the whole margin), the Ross front retreats ≈ 440 km (section: ice base $-200 \dots -280$ m $\rightarrow$ open ocean across four degrees of latitude).

Fix (93a207ee): the DIRECT route. PISM consumes FESOM's own cavity fluxes via `-ocean given: fw→shelfbmassflux ( $\times 1000$ , sign preserved), sst→shelfbtemp`. One model owns the ice-ocean interface. The `-ocean given` machinery existed all along but silently fell back to `th` because its input is a FESOM1.x-era bundle FESOM2 never wrote; `fesom2ice` now synthesizes it (plus three exhumations of never-run 2D-path code: 3f07d7a2, 74671c27, bb9d00c2).

1.3 Bug II — the exchange weights scrambled the coupled ocean (2026-07-14)

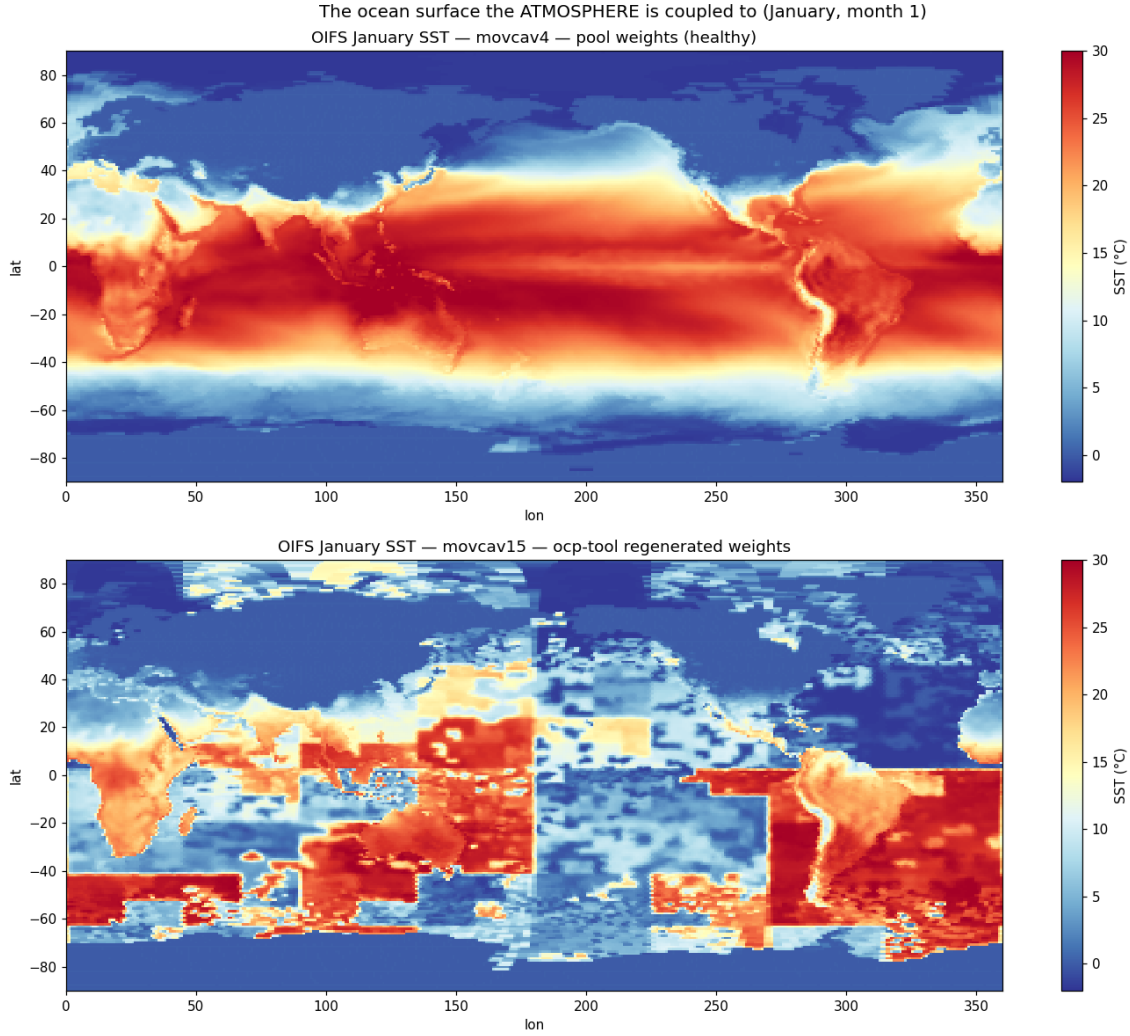


Figure 2: **The picture that settles it.** January SST as OIFS holds it. Top: pool weights — textbook. Bottom: offline-engine weights — a *tile-translocated jigsaw*: tropical blocks at 55°S, polar blocks on the equator, a seam at lon 180 where the node ordering breaks.

FESOM exchanges coupling fields in *partition* (*my_list/rank*) order; the offline weight engine (2026-06-17) built the feom geometry in *nod2d.out* order (median displacement 58°). Every submesh exchange was geographically tile-translocated (Fig. 2): OIFS saw Weddell ice fraction 0.04 while FESOM had 0.95; global sea-ice death followed within months.

Fix (c21fe7c2): `permute_feom_to_runtime.py` reorders the OASIS geometry to runtime order (via `dist_<nproc>/rpart.out`); `validate_feom_weights.py` applies every weight file to real latitudes and **aborts the leg** above 1° median error (nod2d-ordered weights fail at 42–58°, correct ones pass at $\leq 0.025^\circ$).

1.4 Bug III — the OASIS coupling restart was compound-scrambled (2026-07-14)

`remap_oasis_restart.py` gathered in nod2d space, but its input and required output are both *runtime*-ordered: the emitted `rstos.nc` matched *neither* ordering (correlation of `sst_feom` with the

restart’s own surface temperature: +0.15). OIFS’s first coupling interval saw 305 K water under polar air — a one-hour global jigsaw shock that nondeterministically detonated the convection scheme (identical inputs crashed at hour 19, hour 458, or never).

Fix (abe56ee2): un-permute via the old mesh’s `rpart.out` → gather → re-permute via the new mesh’s; correlation now +0.995. Pool `rstos.nc` rebuilt.

1.5 The wave model’s frozen coastline (2026-07-15)

Chunk-3 died deterministically at its first physics step in `voskin_mod.F90:282` squaring WAM’s Stokes drift. WAM has no runtime tiling: its sea-point set is baked into `wam_grid_tables` at preprocessing time and had never been regenerated. A cell that is ocean per the regenerated lsm but land per WAM never receives wave data; the Stokes/Charnock fields are srf-restart-carried, so a *warm* start after a coastline change reads never-written storage (cold starts zero-initialize — why chunk-1 tolerates its 12 mismatch cells and historical lsm changes never hit this). Chunk-3 had 125 mismatch cells. WAM itself was separately exonerated for the Bug-IV crash family (a WAM-off run crashed identically).

Fix: one-time maximal-ocean tables (option A; tolerance patching rejected). WAM’s sea-point set was rebuilt from ETOPO1 with the southern limit extended $78 \rightarrow 85^\circ\text{S}$, then *max-ocean edited*: every cell the FESOM max-mesh can make wet — open ocean or sub-shelf cavity — is forced to sea at its FESOM bottom depth (only ever adding ocean). 1277 cells opened ($29\,555 \rightarrow 30\,832$ sea points), 283 of them south of 55°S : exactly the band a moving coastline walks through (Fig. 3). The tables never change again, wave restarts stay index-compatible across all future mesh changes, and WAM cold-starts once at the switch leg. Known cosmetic debt: the max-ocean edit did not update the obstruction coefficients.

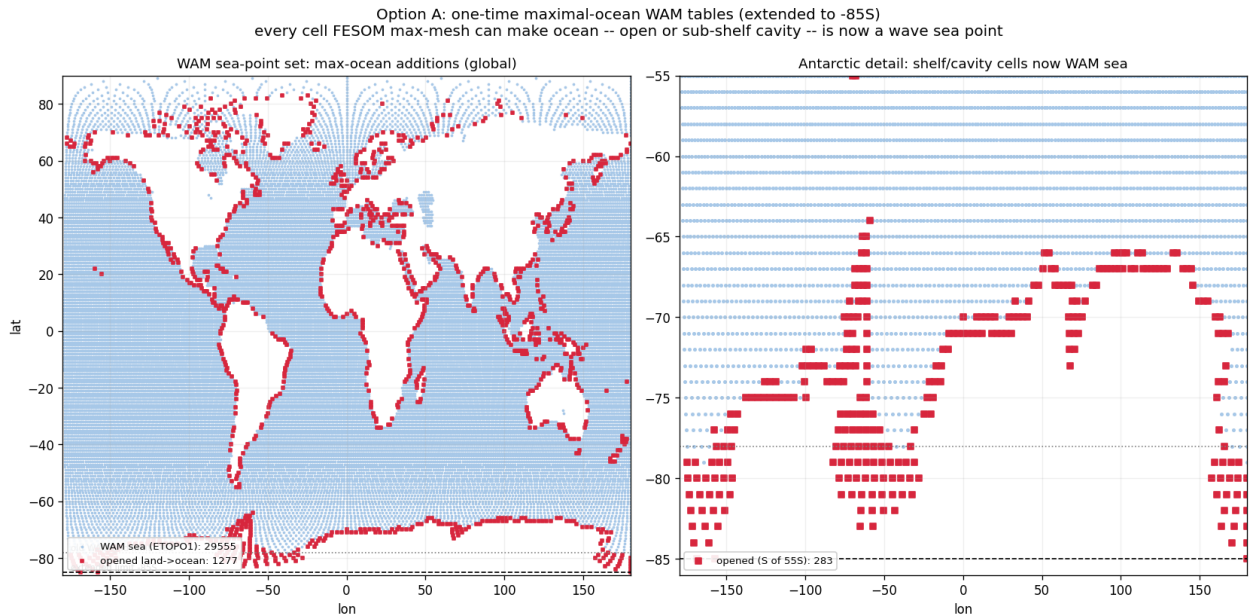


Figure 3: One-time maximal-ocean WAM tables (extended to 85°S). Blue: wave sea points from ETOPO1. Red: cells opened land→ocean because the FESOM max-mesh has a water column there. Pre-registering them makes the `voskin` step-0 crash structurally impossible on any future leg.

1.6 Bug IV — the ice-surface-temperature coupling is architecturally unstable (2026-07-14→16)

This is the mid-run OIFS crash family (`cubasen/vsurf/voskin` floating overflow, random timing, polar/MIZ ranks). It survived three partial fixes before the root was found; the full chain is worth

recording because each layer was real, just not sufficient.

Layer 1: MIZ aliasing of the sent temperature. FESOM sent raw `ist`; ice-free nodes send the seawater freezing point, so the GAUSWGT cell blend in the marginal ice zone was dominated by warm open water: OIFS’s ice tile saw a surface tens of K too warm and returned a strongly negative flux to the fully-iced nodes. Partial fix: concentration-weighted `ist` (`ist·a_ice ÷ ci` on ingest, the EC-Earth/NEMO convention behind `ECE_CPL_NEMO_WEIGHTED_ICE`). Runs survived $3\times$ longer, still crashed.

Layer 2: the FESIM solve has no flux feedback within the coupling interval. FESIM’s skin solve relaxes toward $t^* = T_f + a2ihf \cdot zsniced/con$ with the received flux *frozen* for the hour and no dQ/dT term: under insulation the equilibrium runs away (measured floors 145–183 K; excursions to –954 K in one run). Partial fix: an implicit flux linearization anchored at the transmitted temperature, $Q(t) = a2ihf + \lambda(t_{ref} - t)$ with $\lambda = 4\epsilon\sigma t_{ref}^3 + \lambda_{turb} \approx 20 \text{ W/m}^2/\text{K}$ (the NEMO-family fallback derivative), energy-closed by booking the linearization heat to the ice growth budget (fesom branch `awiesm3-implicit-ice-surftemp`, 38a558b5). Dips became bounded and self-recovering in most runs — but not all.

Layer 3 (the root): the atmosphere’s ice skin was never solved at all. Send-side instrumentation of `A_Q_ice` (term decomposition + tile state at every $|Q| > 1 \text{ kW/m}^2$ event) delivered the verdict:

- The received flux at cold FESOM nodes is **unphysical and anti-braking**: mean -15.8 kW/m^2 below 150 K (100% negative), -4.5 kW at 150–180 K, recurring all year with the MIZ seasonal cycle — at **49°N on 0.1–0.5 m ice with thin snow** (Okhotsk / St. Lawrence). Insulation is irrelevant there; no realistic λ brakes kilowatts.
- At the OIFS send, the generating cells have ice-tile fractions down to 0.04 and **tile skin temperatures of $\sim 204 \text{ K}$ under $\sim 275 \text{ K}$ maritime air**; the sensible + latent terms then reach $+17.8 \text{ kW/m}^2$ per tile area. With weighted-`ist` the tile skins were *still* 189–204 K at substantial tiles — the cold temperature is genuine FESIM state, faithfully communicated.
- The reason the skin can sit 80 K below the air: `vlamsk_mod.F90`, sea-ice tile, with `LNEMOSICOUPL=.true.` (the default): `PLAMSK = 1e10` — **the skin conductivity is set to infinity, pinning the tile skin to the prescribed (remapped FESOM) ice temperature**. The atmosphere’s surface scheme was forbidden from solving its ice surface; it computed fluxes against whatever arrived through the remap.

The loop, in one line: FESIM’s oscillating solve $\rightarrow$ remap $\rightarrow$ *pinned* OIFS skin $\rightarrow$ kW fluxes against an unresolvable surface $\rightarrow$ remap $\rightarrow$ FESIM slams further — a two-model positive feedback closed through the marginal ice zone, metastable enough that identical configurations coin-flipped between completing a year and dying at day 2 (runs `movcav25–31`: raw/weighted $\times \lambda = 3.4/20 \times$ dirty/clean initial state — every combination crash-prone).

Root fix: the coupled slab (§1.8). Stop prescribing the atmosphere’s ice skin altogether; let OIFS solve it implicitly each timestep with conduction through the *coupled* FESOM ice and snow thickness. This removes the loop by construction while keeping the flux honest about the actual ice state. Upstream note: mainline AWI-CM3 carries the same pinned-skin architecture and therefore the same latent (if rarer) crash mode.

1.7 Bug V — iterative coupling never handed over to PISM (2026-07-16)

The `awiesm3` $\rightarrow$ PISM model switch is resolved at `SimulationSetup` init by reading the `chunk_date` file, whose path `esm_runscripts/chunky_parts.py` builds by *raw string concatenation of `general.base_dir`*

before `esm_parser` resolves variables. A `base_dir` containing `${user}` was probed verbatim; the silent `isfile()` miss fell back to “chunk 1, model 1” — awiesm3 self-resubmitted to its own final date and PISM never ran. Proven by log comparison: identical resubmission command lines resolve to `Valid Model Names = ['pism']` under a literal `base_dir` (movcav16 era) and to `['fesom', 'oifs', ...]` under `${user}`.

Fix (d61297e8): `_early_resolve()` expands `${...}` against `general` (environment fallback) in all three chunk-file path builders. Regression-tested against a real chunk file, and validated live: the first genuine awiesm3→PISM handover of the campaign (movcav27), and a full awiesm3→PISM→couple_out→chunk-3-couple_in chain (movcav29).

1.8 The coupled-slab architecture (2026-07-16/17)

The new division of labour: **OIFS owns the ice surface temperature; FESOM owns the ice state.** OIFS’s surface scheme solves the ice-tile skin implicitly within every atmosphere timestep (full radiative + turbulent dQ/dT , finite skin conductivity), with slab conduction through the coupled per-ice thickness and snow depth. FESOM’s albedo still drives the radiation; FESIM’s own `ist` solve remains for its internal state (albedo, melt ponds) but is read by nobody — the feedback loop is cut by construction. The machinery is EC-Earth/KNMI heritage (ICESTATENEMO, 2008; the SI3 branch of `callpar`) — our contribution is wiring the FESOM branch, which lacked only the thickness.

New exchange field. FESOM sends `sit_feom` = effective (grid-mean) ice thickness `m_ice` (o2a collection, `A_Ice_thickness`); `ECE_FESIM_GET_ICE_STATE` divides by the received ice fraction (standard ε -guard) for the per-ice thickness the slab needs, falling back to the climatological 1.5 m only where the cell has no significant ice. Snow thickness (`snt_feom`) was already exchanged and now actually reaches `surfseb` via `SURFL%ZSNTICE`.

The switch set (verified in code, not inferred):

switch	setting	verified effect
LNEMOSICOU	.false.	<code>vlamsk</code> : sea-ice tile skin conductivity becomes <i>finite</i> (land-tile-style lookup) instead of 10^{10} — the skin is genuinely solved by the SEB. The prior default pinned it to the prescribed ice temperature; this pin was the kilowatt engine of Bug IV.
LNEMOLIMTEMP	.false.	disables the hourly overwrite of OIFS’s 4-layer ice temperature (<code>SP_SB YTL</code>) from the FESOM ist ingest; the profile is OIFS-prognostic (ICMGG-initialized, restart-carried). FESOM’s ist send becomes inert.
LNEMOLIMTHK	.true.	activates the <code>callpar</code> fill of <code>ZTHKICE/ZSNTICE</code> from the coupled fields — the skin/top-layer conduction feels the real ice thickness and snow insulation. (The deep 4-layer discretization keeps its fixed depths; the skin conduction is the term that controls winter fluxes.)
LMCCDYNSEAICE	<code>.true.</code>	master enable; false force-resets the whole <code>LNEMOLIM*</code> family (<code>sumcc</code>).
LNEMOLIMALB	<code>.true.</code> (default)	FESOM’s albedo drives OIFS radiation (<code>surfrad_layer</code> → FESIM getter; raw–raw convention, consistent).
ECE_CPL_NEMO_WEIGHTED_ICE	<code>.true.</code> (inert)	only acted inside the disabled ist ingest.
FESOM <code>latm_owns_ist</code>	.true.	FESIM growth budget takes the atmosphere flux directly (<code>Qatmice=-a2ihf</code> , the ECHAM convention); <code>qres/qcon/qlam</code> booking off.

The `sumcc` guard “LNEMOSICOU without LNEMOLIMTEMP does not make sense” correctly enforces that a pinned skin requires a prescribed temperature; our mode flips both off. In one sentence: *before, OIFS’s ice surface was a mirror bolted to whatever FESOM’s oscillating solve sent; now it is a real solved surface conducting through FESOM’s actual ice and snow.*

Validated (movcav33). The first coupled-slab run completed its year crash-free and handed over to PISM. With the instrumentation still armed: kW-scale send events at real ice tiles went from 1410–1930 per run to **zero** (117 residual events are all phantom slivers with ice fraction $\sim 10^{-7}$, a `ZMASK` threshold cosmetic); event tile skins moved from 189–212 K into the physical 247–265 K range (Fig. 4); the fluxes FESOM receives at its coldest nodes are $-63 \dots -80 \text{ W/m}^2$ (textbook polar values, was -10 kW), and FESIM’s internal temperature tracks its anchor to $< 0.1 \text{ K}$.

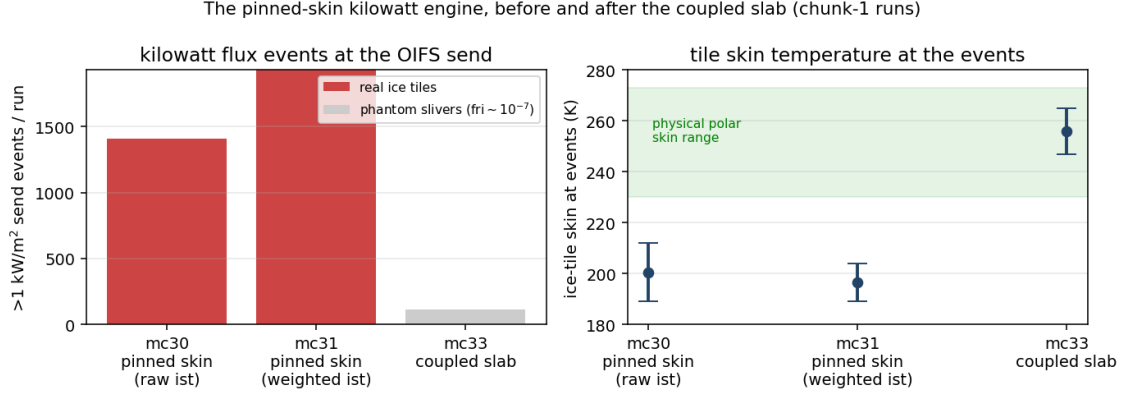


Figure 4: The kilowatt engine, before and after. Left: $>1 \text{ kW/m}^2$ events at the OIFS send per chunk-1 run. Right: the ice-tile skin temperature at those events — pinned-skin runs sit 60–80 K below the physical range; the solved skin cannot.

1.9 First fruits of the clean decade: cavity melt against observations (2026-07-17)

The chunk-1 year that forced the first clean PISM decade, split by sector and set against the satellite record (Rignot et al. 2013; Adusumilli et al. 2020):

sector	observed (m ice/yr)	movcav33 chunk-1 (m w.e./yr)		
		mean	median	nodes
Filchner–Ronne	0.3–0.4	0.89	0.20	1085
Ross	~ 0.1	0.90	0.31	1075
Amundsen–Bellingshausen	14–27 (PIG 16)	7.8	6.4	229
Antarctic Peninsula	0.4–3.8	7.5	4.4	186
Wilkes (Totten sector)	4.7–9.1	8.3	6.4	296
Amery	0.6–0.7	4.0	3.3	128
E. Weddell / Dronning Maud	0.3–0.9	4.4	3.5	287
circum-Antarctic	0.8 ± 0.7	2.84	0.79	3286

The cold/warm dichotomy that dominates the observations is reproduced: the cold-cavity giants sit at 0.2–0.3 m/yr medians while the warm sectors run 4–8, and 13% of the cavity area refreezes (marine ice, as observed under Ronne/Amery). Residual biases: the East Antarctic coastal sectors (Dronning Maud, Amery) run $\sim 5\times$ warm, and the Amundsen runs $\sim 2\times$ cold — both worth revisiting once multi-year coupled statistics exist. Caveats: node statistics (unweighted), m w.e. vs the observations’ mice ($\times 0.92$), one coupled year, and PISM’s initial geometry carries reduced versions of the big shelves. For scale: the jigsaw-era coupling produced 28 m/yr.

1.10 The increment test crashes, twice — and passes (2026-07-17/18)

The first chunk-3 leg — awiesm3 on the PISM-changed 208 438-node mesh, stage 6/6 — absorbs the mesh change and runs for weeks of model time before FESOM dies of an SSH explosion (η jumping $\pm 40 \text{ m}$ in one step at the Antarctic Peninsula tip; day 19–51 depending on MPI reduction order — the state is metastable). The autopsy was cheap for two reasons: the leg costs 3–4 minutes of wall time, and a temporary XIOS file group with `sync_freq="1d"` flushes daily fields to disk as they complete, so even a crashed run leaves full evidence (default XIOS files sync only on close: 51 days of output, zero records). The hunt found *two independent bugs*, stacked at the same place.

Phantom zero-water at draft-changed cavity columns --- George VI / Marguerite Bay
black outlines: elements whose cavity draft the mesh change touched

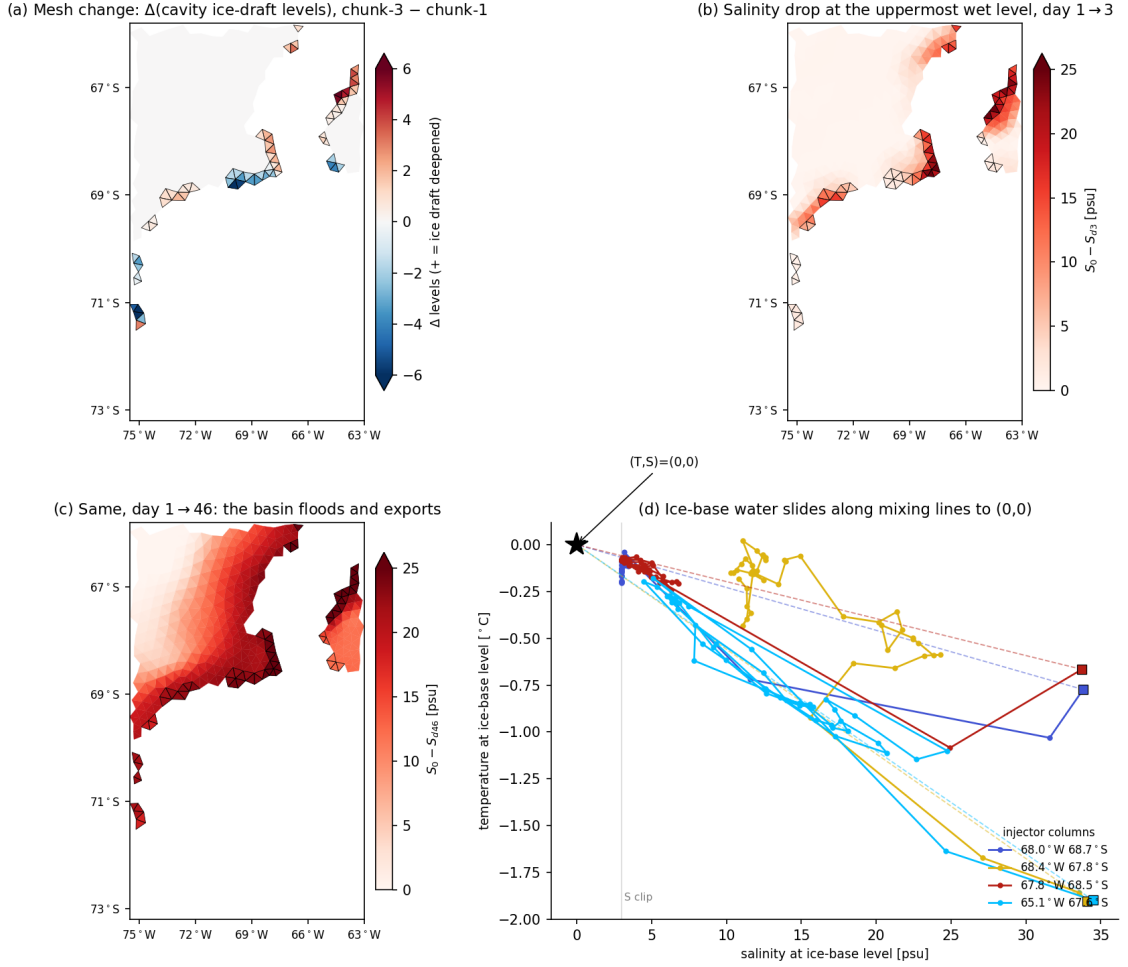


Figure 5: Bug VI seen in one basin (George VI / Marguerite Bay; black outlines = the 73 elements whose cavity draft the mesh change touched). (a) The mesh change: ± 1 –6 cavity levels. (b) By day 3 the uppermost-wet-level salinity has dropped up to 25 psu *exactly on the outlined elements*. (c) By day 46 the basin has flooded and exports along the coast. (d) The fingerprint: (S, T) at the ice-base level slides along the mixing lines toward $(0, 0)$ with *identical* T - and S -replacement fractions — not a flux (surface freshwater is $50\times$ too small and negative), but a *multiplicative drain*: both tracers lose the same fraction of their value every step.

Bug VI — the layer-thickness mismatch drain

The remapped restart carries `hnode` inherited from the old mesh's `zstar` state (e.g. 9.964 m where a formerly ice-free column scaled its layers with η). At runtime, cavity columns and their open-ocean edge ring (`ulevels_nod2D_max > 1`) keep layer thicknesses *fixed at the mesh nominal* (10.000 m) and are excluded from the per-step `hnode ↔ hnode_new` sync — so the mismatch is immortal. FESOM's tracer T^* update applies values $\cdot (\text{hnode} - \text{hnode_new}) / \text{hnode_new}$ every step: a -0.36% /step multiplicative drain on T and S . Per-operator probes at a Getz injector column measured -0.1226 psu/step against a predicted $33.64 \times 0.0036 = -0.1224$; an unchanged FRIS control column measured exactly zero. At free-surface nodes next to the cavity edge the drain even *accelerates*: η grows, the nominal-vs-restart gap grows with it. **Fix** (FESOM, `restart_thickness_ale`): after the restart read, force `hnode` at the whole excluded class back to the fixed nominal and rebuild the depth arrays. Validated by

intervention: the drain vanished ($+0.0002/\text{step}$), day-1 fields healthy.

Bug VII — the drowned island

With Bug VI fixed the run survived longer but still exploded at the Peninsula tip, η climbing a steady $+0.11\text{m/day}$. The reason is geometric, not numeric (Figs. 7, 6): the ice2fesom regrid remapped PISM’s *categorical* mask **bilinearly** — a few-cell island surrounded by mask-4 ocean smears to $\approx 3.x$, rounds to ocean, and is deleted from the submesh. PISM’s own bed there is $+38$ to $+218\text{m}$ above sea level: bedrock island, regardless of ice. (The island entered the campaign *bare* — the spinup state carries 0.0m of ice on all ten cells; a thin $0.1\text{--}0.4\text{m}$ cap has been *growing* during the coupled decades. A second, independent deletion path was found while examining it: PISM encodes ice-free bedrock as $\text{mask}=0$, which the node classifier silently defaulted to ocean; fixed — bare bedrock above sea level is land.) The ocean then drives a runaway coastal jet where a coastline should be. **Fix** (ice2fesom): remap the categorical mask nearest-neighbour; rebuild the $\text{mask}=5$ outside-domain marker (which `remapnn` extrapolation would erase) from a bilinear domain indicator. Continuous fields stay bilinear.

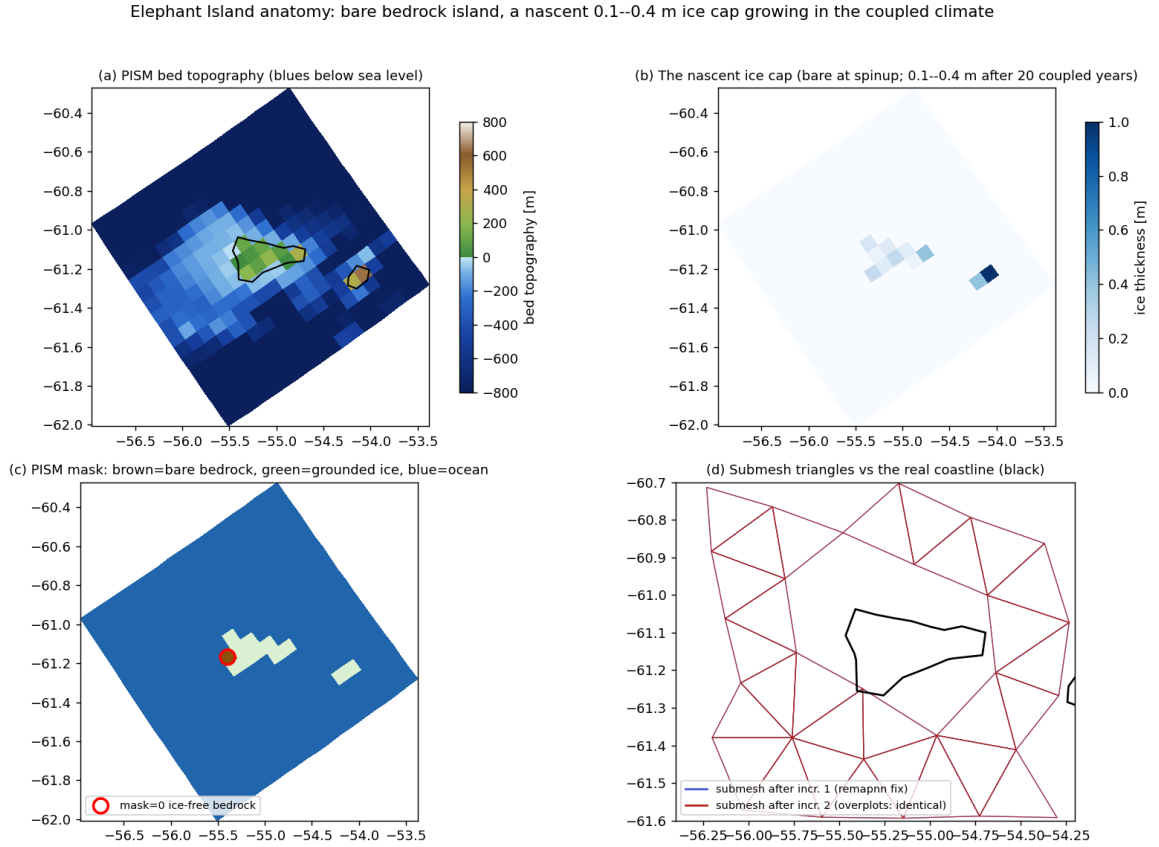


Figure 6: Elephant Island anatomy. (a) PISM bed topography, ETOPO-style colors: a genuine bedrock island ($+4$ to $+297\text{m}$) on its shallow platform, Clarence’s peak to the southeast. (b) The *nascent* ice cap: bare at spinup, $0.1\text{--}0.4\text{m}$ grown over 20 coupled years (DEBM accumulates here). (c) The PISM mask; the red circle is the single $\text{mask}=0$ (ice-free bedrock) cell that exposed the classifier hole. (d) The submesh renders the island identically after increments 1 and 2 — the nearest-neighbour fix holds.

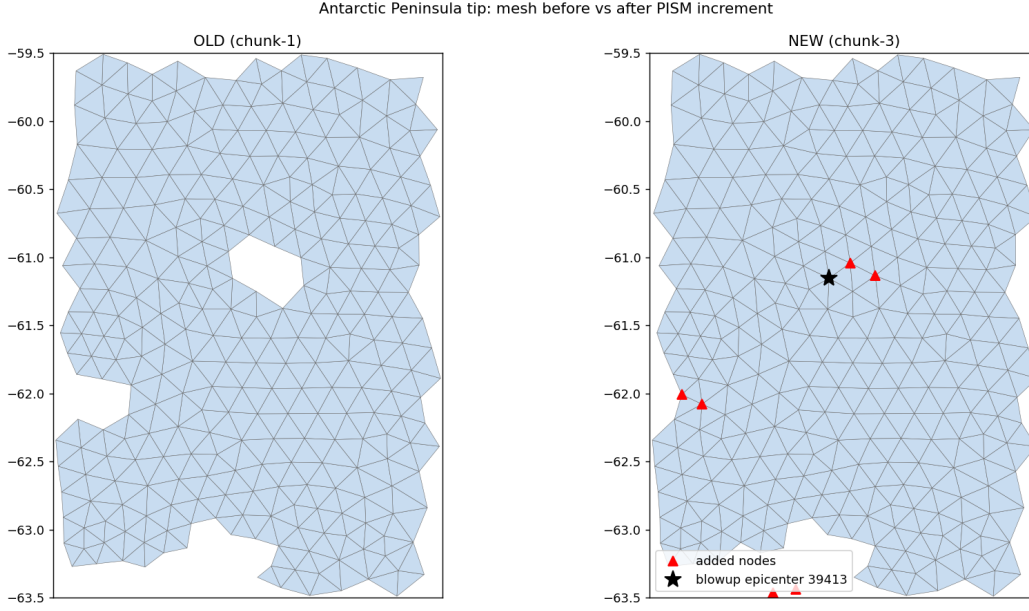


Figure 7: Bug VII. Left: the chunk-1 mesh has Elephant Island (the hole at 55.4°W , 61.1°S) and the Clarence/Gibbs notch. Right: after the PISM decade the bilinear mask remap has rounded both islands to ocean — the added nodes (red) fill them in, and the blowup epicenter (star) sits exactly on the drowned island.

Ruled out by measurement (each was a plausible suspect): surface freshwater fluxes; OIFS runoff/calving and the ECE_LANDICE snow cap; the OASIS `feom` masks; the remapped T/S state (freezing cap ran: 1137 columns); mesh-file level-range consistency (zero violations); the zero-filled dry restart levels (padding them — kept as a permanent hygiene pass — reproduced the crash bitwise); and the vertical-advection top boundary (cavity guards in both the implicit and explicit schemes changed nothing — see §4). The couple_in warning about a missing `cap_new_cavity_temp.py` is a red herring: the freezing cap runs as Fortran inside the remap binary.

The increment test passes

With both fixes, `movcav33` chunk-3 completed the full year 1901 on the changed mesh (25 min wall, no blowup; the Peninsula node’s SSH wanders seasonally instead of pumping) and handed over to the chunk-4 PISM decade unattended — **stage 6/6 closed**. The long run `movcav34` (10 awiesm3 years / 100 PISM years) then repeated chunks 1–4 cleanly on the *minimal* build (hnode fix only; the advection guards were reverted after being falsified). Fixes are committed: FESOM `awiesm3-implicit-ice-surftemp`, `esm_tools development_interactiv_mesh_awiesm3`.

The next problem: the second increment and the hygiene overreach (2026-07-18)

`movcav34`’s chunk-5 (year 1902, after 20 PISM years) blew up at day 16 at the western Ross front, where the second increment opened cavity columns wholesale (ulev 13 $\rightarrow$ 1) amid a 506-node ice advance. The new coastline there was structurally rotten: 24 nodes within 1.5° of the epicenter held by ≤ 3 elements, including an 18-levels-under-ice cavity column held by exactly two triangles. A *coastline-hygiene sweep* was added to the submesh tool (iteratively remove ice-domain nodes held by ≤ 2 elements) and the chain then ran unattended through year 1911 and 120 PISM years — eleven further increments without a FESOM blowup, ending only in a recurrence of the old OIFS `voskin` overflow at chunk 13.

The overreach. At wrap-up the evolution diagnostics exposed the truth (Figs. 8, 9): the

hygiene sweep had *cascaded through the narrow cavity bands* — fringing shelves and cavity margins are exactly the few-element-wide structures the criterion eats rim-inward — and every post-1902 submesh carries only 14–111 cavity nodes instead of ~ 2600 . The node classifier was healthy (2223 shelf nodes flagged); the lake floodfill removed only 35 elements; the sweep itself did the damage. **The eleven “stable” legs ran with the ice-shelf cavities amputated — stability by amputation, and every result from movcav34 after chunk 4 is TAIKED.** The sweep was reverted the same day (commit ca0641b2 reverts 680daa55).

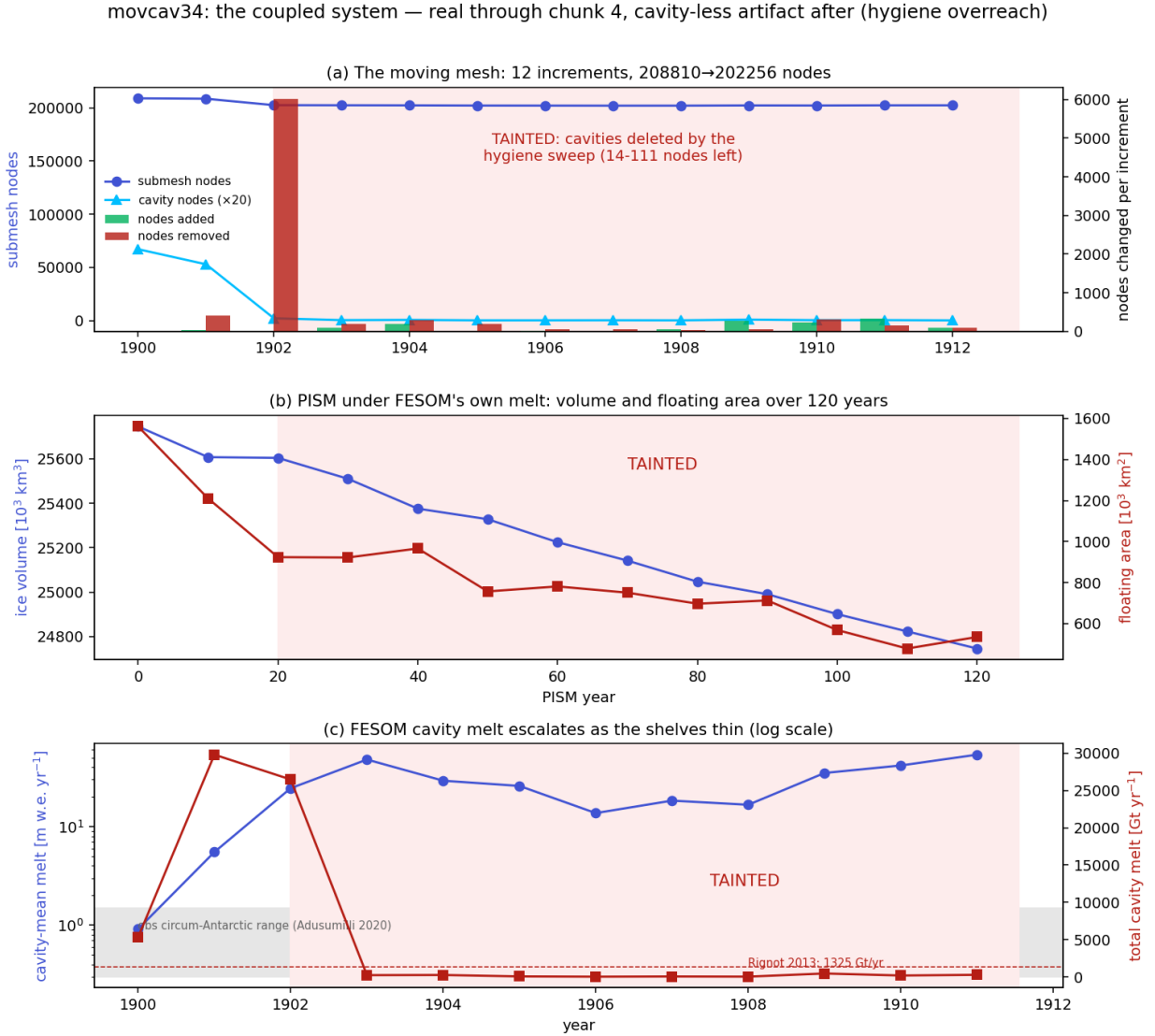


Figure 8: The movcav34 production run, honest version. (a) Submesh and cavity-node counts per increment: the collapse from 2647 to 111 cavity nodes at the 1902 mesh is the hygiene amputation. (b) PISM volume and floating area over 120 years: the decline is real coupled melt through year 20, then continues under forcing computed from an essentially cavity-less ocean. (c) Cavity-mean and total melt per leg (log scale): the escalation beyond ~ 1902 samples only the few surviving cavity slivers. Red shading = tainted.

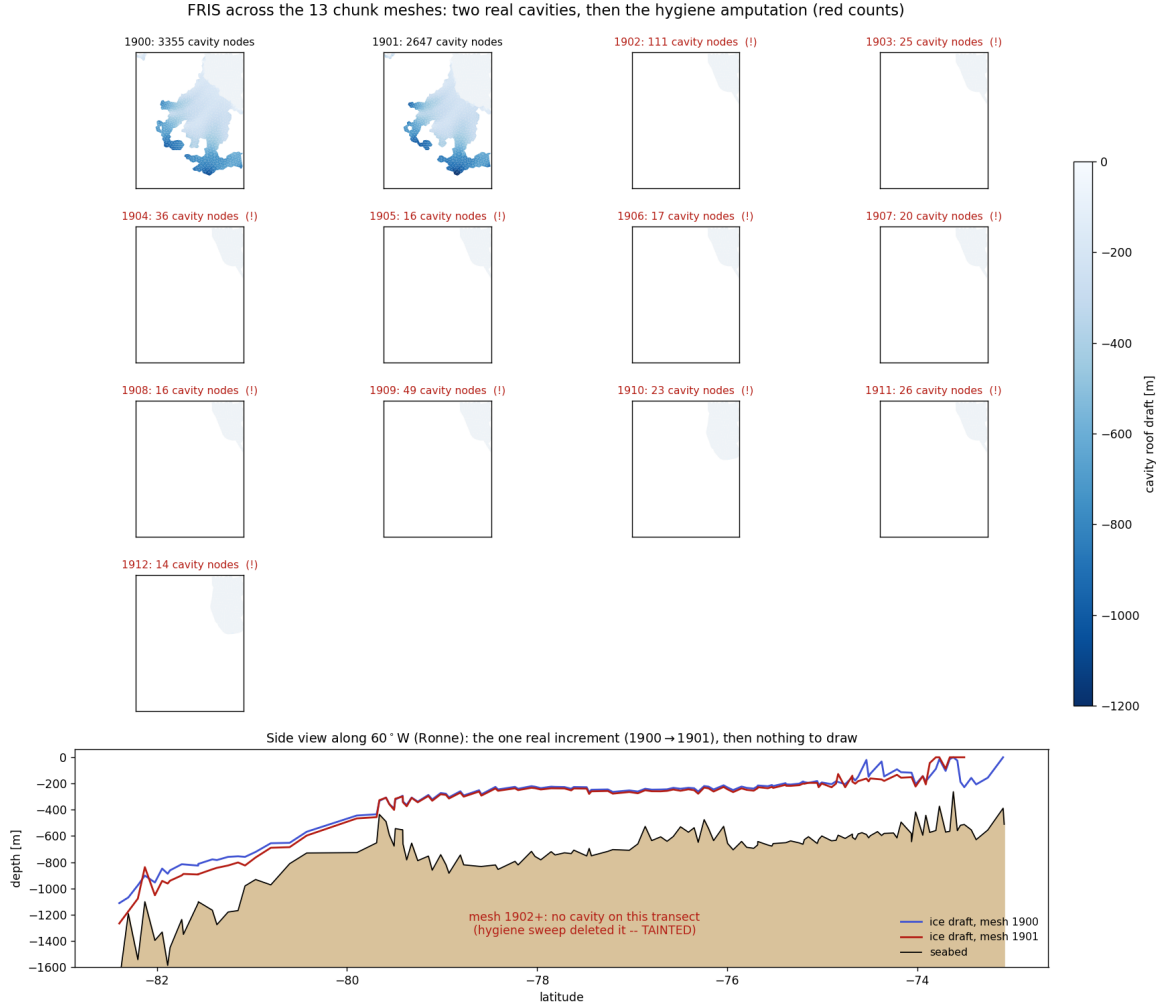


Figure 9: FRIS across the thirteen chunk meshes. Top grid: cavity elements coloured by roof draft — two real cavities (1900, 1901), then eleven empty frames with red node counts. Bottom: ice draft along 60°W for the one genuine increment (1900→1901; the southern draft deepens by 50–100 m), after which there is nothing left to draw.

The problem to solve next (Figs. 8, 9 are its statement): make the mesh increments survivable *without* destroying the cavities. What is genuinely proven and stays: the full cycle awiesm3 → PISM → awiesm3-on-a-changed-mesh works end-to-end with real cavities (movcav33 chunk-3 + handover; movcav34 chunks 1–4 reproduce it on the minimal build). The work queue, in order: (i) a cavity-preserving fragile-sliver treatment — removal must not disconnect or cascade through the few-element-wide cavity bands (guard on the shelf flag, a connectivity-preserving criterion, plus a safety valve that aborts the leg if the submesh cavity count drops far below the classifier’s flag count); (ii) rewind movcav34 to chunk-5 and re-run the increments with real cavities; (iii) the recurring OIFS voskin coastal overflow; (iv) the iterative-coupling loop ignores `final_date`; (v) the escalating warm-bias melt that drives PISM’s fast geometry increments in the first place. Items (i) and (v) are worked below.

1.11 The coastline is not the problem — the increment is (2026-07-18)

The Ross-front blowup looked like a fragile coastline (24 nodes held by ≤ 3 elements within 1.5° of the epicentre, including an 18-level cavity sliver on two triangles), so the first response was a hygiene sweep removing ice-domain nodes held by ≤ 2 elements. It ran the chain to year 1911 — but by *amputating the cavities* (Bug “VIII”, above). An offline harness on the saved 1902 flag files then settled the question by building submesh variants without submitting jobs:

- **Fragility is normal, not pathological.** The meshes that ran clean full years carry the *same* fragile census as the crashed one: pool 3470 nodes with ≤ 2 elements (429 in cavities), submesh_1901 3448 (378), versus 1902’s 3444. A control that kills the hypothesis: if ≤ 2 -element slivers were fatal, 1901 would have died too.
- **The increment is the violence.** Measured as per-node cavity-roof change against the previous submesh: increment 1 (pool→1901) max $|\Delta\text{draft}|$ 451 m, mean 0.96 m; increment 2 (1901→1902) max **1196 m**, mean **8.7 m** — $9\times$ the mean, opening 13 vertical levels at the epicentre in one decade beside a 506-node grounding advance. This is downstream of item (v): the warm-bias melt escalates (cavity-mean $0.9\rightarrow 5.5\rightarrow 24$ m/yr), so each decade’s geometry step is larger than the last.

Fixes committed. (a) A *cavity-amputation abort guard*: if the reduced submesh retains $< 50\%$ of the classifier’s flagged cavity nodes, the leg fails loudly instead of running a formally-valid but cavity-less ocean (the tripwire that would have caught Bug VIII in one couple_in instead of eleven tainted chunks). (b) The PISM mask-value 0 (MASK_ICE_FREE_BEDROCK) is now classified *land*: a second, independent island-deletion path (bare bedrock defaulting to ocean) that would have re-drowned Elephant Island once its nascent cap finishes melting. **Rejected:** a per-increment cavity-roof *rate limiter* (clamp the roof to within 75 m of the previous leg). It worked offline (retained 2221 cavity nodes vs 1864 unclamped) but was *rejected on physics grounds*: it makes the ocean’s cavity geometry deliberately lag the ice sheet’s, a coupler-imposed inconsistency between components. The increment must be tamed at its source (the melt, next) or by numerical treatment of the post-increment transient — not by distorting the coupled geometry.

1.12 Why the cavity melt is high and rising, and how to slow it (2026-07-18)

Item (v), the disease under the increments: the PI-control cavity melt runs at ~ 5.5 m/yr area-averaged against an observed circum-Antarctic ~ 0.8 m/yr, and *escalates* ($0.9\rightarrow 5.5$ m/yr over 1900→1901 on real cavities). Diagnosed from the monthly `temp/salt/fw/unod/vnod` output (Fig. 10); the cause is not where the first pass guessed.

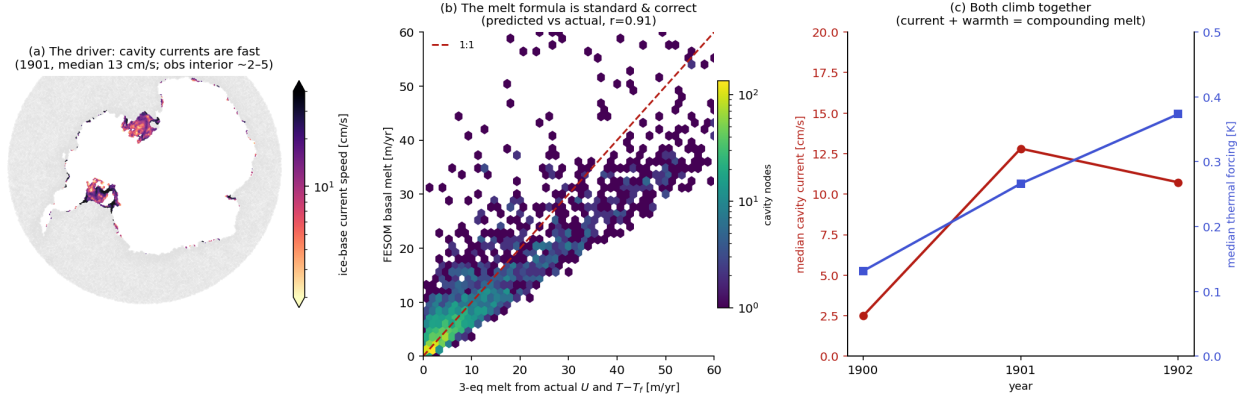


Figure 10: Why cavity melt is high and rising. (a) Cavity ice-base current speed (1901): median 13 cm/s, mean 18, 61% of nodes above 10 cm/s — fast, concentrated at the Amundsen/Bellingshausen shelves and the Ross/Filchner ice fronts (observed cavity interiors are a few cm/s). (b) The melt *formula* is standard and correct: fed FESOM’s actual currents and thermal forcing, the three-equation scheme reproduces the model’s own melt (predicted vs actual $r = 0.91$). (c) Both drivers climb together — thermal forcing (median $+0.13 \rightarrow +0.37$ K, warm-water intrusion) and current speed — so the melt compounds.

The chain, measured. Thermal forcing $T - T_f$ at the ice-base level is *near-realistic* (area-weighted $+0.14$ K; median $+0.13$ in 1900 rising to $+0.37$ in 1902). The Holland & Jenkins three-equation scheme in `cavity_param.F90` is standard (drag $a_k = 2.5 \times 10^{-3}$, $\gamma_T \approx 10^{-4}$ m/s), and fed the model’s own currents and thermal forcing it *reproduces the melt* ($r = 0.91$, 18 vs 16 m/yr): the formula is not miscalibrated. By elimination — melt $7\times$ too high, formula validated, temperature realistic — the excess lives in the **cavity currents**: median 13 cm/s where ~ 2 cm/s would give the observed melt, i.e. $\sim 6\times$ too fast. Because the exchange velocity is $\gamma_T \propto |U|$, that fast circulation is exactly what inflates the melt; and as the shelves thin the currents speed up further — the escalation.

How to slow it without killing the climate response. The design constraint is to dissipate *momentum* with a flow-dependent operator, never to cap the diagnostic (velocity, γ_T , or melt) — a quadratic drag still lets the current find a slower equilibrium that shifts with forcing, whereas a cap freezes the response. The ice base is *not* free-slip: `cavity_momentum_fluxes` already applies quadratic drag $-\rho C_d |U|U$ at `nzmin`, correctly ordered (cavity elements skip the atmospheric-stress overwrite, then receive the drag). But it borrows the *seabed* coefficient $C_d = 2.5 \times 10^{-3}$ with the developer’s own flag in the source (“*need to check the sensitivity to the drag coefficient*”). The plan: split it into a separate, tunable `C_d_cavity` (default = C_d) and raise it — physically justified, since ice-shelf bases are hydraulically rougher than a flat seabed (basal crevasses, terraces, keels), and dedicated cavity models treat ice-base drag as a distinct tunable $\gtrsim$ seabed. Being quadratic it preserves the velocity’s climate sensitivity. Complementary flow-dependent levers if drag alone is insufficient: a vertical-viscosity floor in the cavity boundary layer, or Smagorinsky horizontal viscosity in cavities. *Not* to be done: raising the global C_d (over-brakes the open-ocean bottom boundary layer everywhere). Validation target: melt toward ~ 0.8 m/yr *while preserving* the spatial melt-forcing pattern ($r = 0.81$) and the warm-Amundsen/cold-FRIS contrast — brake the amplitude, not the geography. Caveat: part of the $2.5 \rightarrow 13$ cm/s jump over 1900 $\rightarrow$ 1901 is plausibly spin-up plus the increment shock, so retune against a settled multi-year state, not a transient.

On bias correction (a design question raised alongside): observation-anomaly (“delta”) ocean forcing is standard for stand-alone ice-sheet runs, and even cavity-resolving projections bias-correct their *atmospheric* forcing (Naughten et al. 2018 force FESOM with CMIP anomalies on an ERA-Interim climatology, control on reanalysis) — but it would not cure this run: the thermal forcing

here is already near-realistic, so correcting T/S addresses a problem we mostly do not have. The excess is in the *circulation*, upstream of both the melt formula and the water it is fed.

2 State of the cycle and of the climate (2026-07-18)

2.1 What is proven

- **Chunk-1 is quantitatively healthy and, under the coupled slab, crash-free:** January SST physical, zero translocated cells; OIFS Weddell September ci 0.61/0.99 (was 0.04); both-hemisphere sea-ice annual cycles; per-basin cavity melt in the observed regimes (§1.9); kilowatt flux events at real ice tiles: zero.
- **The workflow hands over in both directions** (Bug V fixed): awiesm3 $\rightarrow$ PISM $\rightarrow$ couple_out $\rightarrow$ chunk-3 couple_in runs unattended.
- **The PISM decade runs on FESOM's own melt** (-ocean given, 2026-07-17): real output, and a *gentle* geometry response — 290 fully-opened nodes vs 1399 in the Bug-I era, mean $|\Delta\text{draft}|$ 1.0 m, submesh 208 810 $\rightarrow$ 208 438 nodes (Fig. pism_change_mc33).
- **The live mesh change works:** weights regenerated and runtime-order-validated ($\leq 0.024^\circ$), restarts remapped through the rpart sandwich, ICMGG/plit regenerated.
- **The increment test passes end-to-end** (2026-07-17/18): chunk-3 runs its full year on the changed mesh and hands to chunk-4 PISM unattended; movcav34 (10 awiesm3 years / 100 PISM years) repeats chunks 1–4 cleanly on the minimal committed build. The second increment's Ross-front blowup (§1.10) is the current front.
- **The high, rising cavity melt is diagnosed** (§1.12): near-realistic thermal forcing (+0.14 K) through a standard, validated melt scheme ($r = 0.91$), driven by cavity currents $\sim 6\times$ too fast (median 13 cm/s). Fix planned: a separate tunable ice-base drag C_{d_cavity} . Open.
- ~~The chunk-3 dilution enters through the vertical-advection top boundary (zero-tracer inflow ratchet / material-interface flux).~~ Cavity guards in both the implicit (`adv_tra_vert_impl`) and explicit (`upw1`, `qr4c`) schemes changed the crash bitwise-nothing; the guards were reverted. The identical-fraction $(T, S) \rightarrow (0, 0)$ signature belongs to the *multiplicative* `hnode` mismatch drain (Bug VI).
- ~~The dilution is unclosed column continuity: remnant $\nabla \cdot (u h)$ at pinned- η cavity columns drains tracer content at fixed volume.~~ The divergence and ice-base w are symptoms of the same `hnode` inconsistency, not an independent leak; with the restart `hnode` reset the columns are quiet.

2.2 Open item: the Southern-Hemisphere summer melt-back

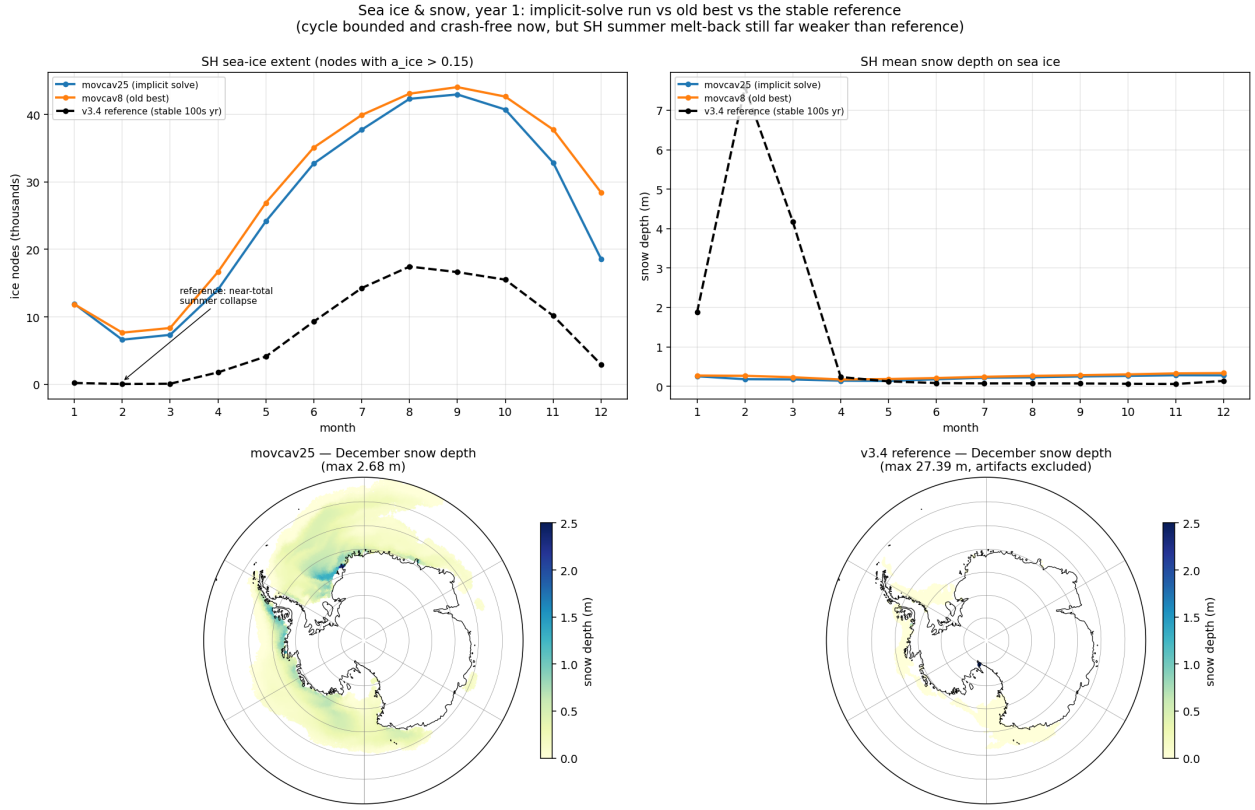


Figure 11: Year-1 sea ice and snow: implicit-solve run (blue) vs the movcav16-era state (orange) vs the 500-year-stable v3.4 reference (black dashed). Our runs track each other — the crash fixes did not degrade climate — but the reference collapses its SH ice almost completely each summer while ours retains $\sim 7k$ nodes, and its winter pack carries 0.03–0.08 m of snow against our 0.2–0.3 m. Bottom maps (same scale): December snow depth — our thick band hugs the Antarctic coast; the reference is nearly bare.

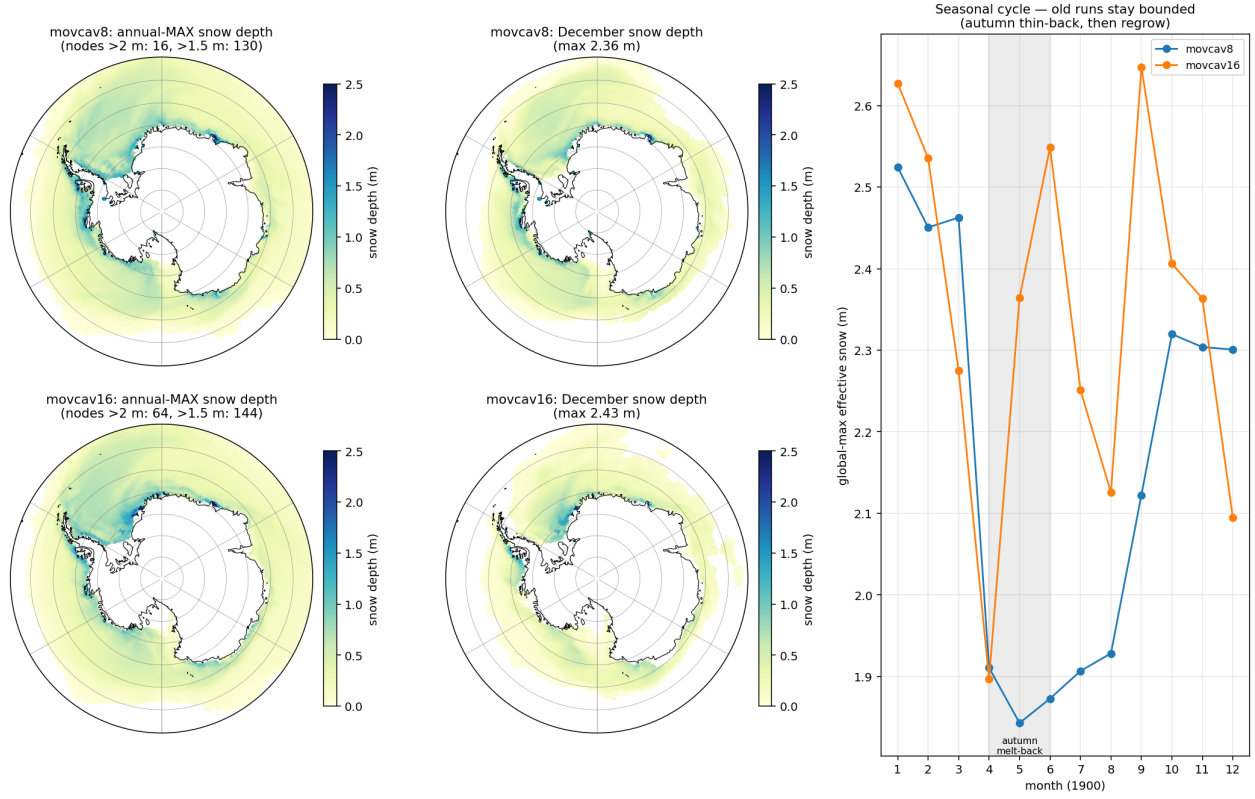


Figure 12: The extent of the coastal pile-up in the (pre-fix) movcav runs: annual-max and December snow depth, and the seasonal cycle. The thick snow is a band hugging the ice-shelf front (eastern Weddell/Fimbul, Amundsen), bounded at 2–2.6 m by an autumn melt-back. Later runs that lost this melt-back drifted monotonically to 3.5 m — the insulation that raised Bug IV’s loop gain.

The ice-thermodynamics code is byte-identical to the v3.4 reference; the difference is state and forcing. Ours keeps $\sim 2.5\times$ the reference’s winter SH extent and only a $\sim 6\times$ summer retreat where the reference manages $\sim 200\times$ (Fig. 11), and piles 2–3.5 m of snow against the Antarctic ice-shelf coast (Fig. 12). Two mitigations exist: a *clean chunk-1 ice initial state* (reference year-1849 fields indexed onto the submesh via `map_nod.out`, 7 Ronne artifact nodes hybridized; snow max 0.81 m — in use since movcav29), and the coupled slab itself (the pinned-skin era’s fictitious equilibria drove a spurious congelation-growth term). Whether these restore the reference’s melt-back is the first climate question for the post-slab runs; meltponds were checked and are *not* the SH lever (the `hsno>hs1` lockout makes old and corrected schemes identical there).

3 Secondary bugs, fixed along the way

Bug	Fix	Commit
Remap extrapolated unboundedly below the old seabed (manufactured -45°C) when columns deepen	take water from the nearest old node wet at that depth	eef4f6db
Sub-ice η/hbar kept open-ocean values at newly-sub-ice nodes	zeroed (real bug, not the cure)	b17fb5c8

Bug	Fix	Commit
First mesh-change leg read the mother mesh as cavity-free (cavity_nlvls.out missing from the mother_as_old links) $\Rightarrow$ zeros in newly-wet levels, mstep-1 blowup	link the cavity files	functions
Same-chunk couple_in rerun promoted its own submesh to previous_submesh (old = new $\Rightarrow$ zeros again)	rerun guard	b9c1e370
OASIS GSSPOS conservation blowup on near-zero target residuals	gauswgt_gsmart transform	3e94d4da
namcouple feom dimension never actually patched (env var from another subjob's shell)	fixed variable passing	3cf29e9e
fesom2ice loaded the static full mesh against submesh output	load the run mesh	63cfcc91
previous_submesh read but never created	created; plus restart size guard	3b44a704
Failed coupling functions did not stop the workflow (backgrounded, observe saw no exit code)	.coupling_failed sentinel honoured by observe	7953585b
rstos size guard read the wrong dimension	fixed	61c8ed8a
ocp-tool flipped a coastline cell's mask+soil only, leaving an inconsistent surface column	masked nearest-neighbour rebuild of the full column	ocp-tool PR #51

The recurring pattern: the guards were right and the plumbing around them defeated them — a correct refusal followed by an unconditional continue is worse than no guard at all.

4 Falsified claims (kept, struck through)

- ~~Geostrophically-unbalanced remapped state; initialize changed cells in thermal-wind balance.~~ Failed its own offline gate (discrete ∇p noisier than NN fill); ships opt-in only, off by default.
- ~~The sub-ice ssh BC violation is the blowup.~~ Real bug, fixed, run still died.
- ~~The per-leg PISM increment is gentle (≈ 22 m mean).~~ The mean was diluted by the 98% unchanged domain; the distribution is wildly skewed (max 923 m, 1399 fully-opened nodes).
- ~~The salt plume parameterization being off explains the warm cavity.~~ The spp-off cavity is the *coldest* in the CORE3 reference runs; the warm cavity was Bug II.
- ~~The native ICMGG causes the mid-run crash family.~~ The step-0 swap tests proved the native, coastline-following surface set is *required* on the submesh (62 difference-zone cells cannot initialize otherwise); the mid-run family was Bug IV.
- ~~Weighted-ist cures the crash family.~~ Necessary hygiene for the o2a blend, not sufficient: the pinned OIFS skin regenerated the instability from bounded inputs.
- ~~The insulation/snow pile-up is the crash root.~~ It raises the loop gain (and is a real climate bias, §2), but the kW events occur on *thin* ice at 49°N; the clean-ini run crashed identically.

8. ~~Fragile ≤ 2 -element coastal slivers cause the Ross-front blowup.~~ A control census killed it: the meshes that ran clean full years carry the same ~ 3450 -node fragile count as the crashed one. The cause is increment magnitude, not coastline fragility (§1.11).
9. ~~The cavity melt gain is $\sim 10\times$ too steep (39 vs 1–4 m/yr/K).~~ The 1–4 figure is an integrated-shelf number, not the local exchange sensitivity; the standard three-equation scheme gives ~ 40 m/yr/K locally, so the measured 39 is correct. The melt formula is standard and reproduces the model’s melt ($r = 0.91$); the excess is in the cavity *currents* (§1.12).

5 Method notes

1. **Cheap controls first.** The cold start (geometry vs state), running the fix, plotting the distribution instead of the mean — each killed a confident claim in minutes.
2. **Measurements from a corrupted run are worse than none** (Bug II poisoned Bug I’s diagnosis for a week).
3. **“Same code as the stable reference” localizes nothing by itself** — the reference ran the same FESIM solve for 500 years; the difference was the state and, one layer deeper, a switch default that pinned a skin nobody remembered was pinned.
4. **Instrument both ends of a coupling before theorizing about either.** Receive-side logging (ICEDBG) proved the flux was corrupt; send-side decomposition (SENDDBG) named the term and the tile state in its first twelve lines. The two probes cost one rebuild each.
5. **Beware silent fallbacks** (Bug V’s `isfile()` miss; the `-ocean` given silent fallback to `th`). A fallback on a nonexistent path should say so.

6 Assets

- Send/receive flux instrumentation: SENDDBG (OIFS `A_Q_ice` decomposition + tile state), ICEDBG (FESIM received flux + state at cold nodes), CPLDBG (OIFS ingest bounds) — all log-only, strip before merging.
- `oasis_expout_o2a` runscript flag: every o2a exchange dumped to netcdf ($\sim 20\times$ slowdown; crash-window probes only).
- Clean chunk-1 ice initial state (`prep/clean_ice_ini`): reference year-1849 ice fields indexed-mapped to the submesh; `map_nod.out` makes such remaps exact.
- A 2-minute offline reproducer (standalone FESOM on the PISM-cavity submesh) and a stable full-mesh control.
- `validate_feom_weights.py`: aborts any leg whose exchange weights are in the wrong node ordering — would have caught Bug II on day one.
- `verify_balance.py`: offline balance/stability gate for remapped restarts.
- Chunk-1 artifacts pooled at `input/fesom2/pism_cavity_ini/` (login-node launches).